

The density advantage in FORCE-trained motor RNNs disappears under a matched plasticity budget

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Abstract

We studied rate recurrent neural networks (RNNs) trained with FORCE-style recursive least squares (RLS) on six-target centre-out reaching, following Feulner & Clopath [1]. Recurrent connection density was varied over $p \in \{0.05, 0.10, 0.20, 0.40\}$, with variance-scaled weights and paired seeds. With all edges trainable, denser networks perform clearly better: the contrast H1 (sparse NMSE minus mean denser NMSE) is $+0.190$ (seed-bootstrap 95% CI $[0.154, 0.229]$), positive in 16/16 network seeds at $N = 200$. When every density receives the same number of trainable edges, the advantage disappears: the per-seed drop in H1 is $+0.231$ $[0.193, 0.267]$, positive in 16/16 seeds. This holds up to 200 training trials, under spectral-radius matching (16/16), at $N = 400$ (8/8), at $N = 100$ after 200 trials, and when the budget is a random subset of each dense mask (8/8). An earlier 8-seed analysis suggested that the sign *reverses* under the equal budget (-0.090 $[-0.113, -0.068]$). With 16 seeds this residual is about half as large, and with matched spectral radii it is not distinguishable from zero. A factorial ablation detects no density component in it, although it cannot exclude one of the same size; it is more consistent with a cost of the frozen recurrent input that such a control introduces. In this model the density advantage tracks the number of trainable recurrent edges, not structural density.

1 Introduction

Recurrent connectivity in cortex is sparse. Network models of motor learning usually fix the connection probability: Feulner & Clopath [1] use 10%. Connectivity constrains the low-dimensional activity a circuit can express [2], and sparse random reservoirs can compute well even with their recurrent weights frozen [3]. Recent studies of trained networks report that sparse, structured connectivity can speed learning: locality-masked sparse RNNs with an optimal sparsity train faster and more data-efficiently than fully connected ones on a large cognitive task battery [4], and in large recurrent networks with cortex-like constraints sparse connectivity facilitates time- and data-efficient learning [5].

Sparse, variance-scaled random networks keep the overall recurrent input statistics of dense ones, so one might expect density to matter little once the weights are rescaled. With a learning rule that updates every existing edge, such as FORCE/RLS [6], density changes two things at once: (i) the structure of the recurrent graph, and (ii) the number of weights the rule can change, which scales as pN^2 .

Our Neuromatch Academy project first observed that denser networks learned the reaching task better when every existing edge was trainable. We then asked whether this survives when the number of trainable edges is held fixed. The first control suggested a clean sign reversal. This note reports the robustness checks we ran before trusting that reversal, and what is left of it.

Contribution. A small, fully reproducible case study (paired seeds, held-out evaluation with a fixed decoder, exact identity checks between arms) showing that (1) the density advantage in a FORCE-trained motor RNN is a plasticity-budget effect; and (2) the apparent reversal under a matched budget is small and not robust to gain matching. A factorial ablation detects no density component in it; it is more consistent with a cost of the frozen edges that such a control necessarily introduces. We make no claim about biological circuits.

2 Methods

2.1 Model and task

We use the rate RNN of Feulner & Clopath [1] as adapted in the Neuromatch Academy motor-RNN project template.

Dynamics. N units with state r and rate $z = \tanh(r)$ evolve as $\tau \dot{r} = -r + Wz + W_{\text{in}}u$, with $\tau = 0.1$ s, Euler step $dt = 0.01$ s and 2 s trials.

Task. A 0.2 s one-hot cue u selects one of six centre-out targets. The network must then produce a constant 2-D velocity (magnitude 0.2) through a fixed random linear decoder D ($2 \times N$, Frobenius norm 0.04).

Learning. Recurrent weights are trained online with per-neuron RLS [6] ($P_0 = I/20$; an update every second time step after the cue). The decoder error is assigned to neurons through the pseudo-inverse D^+ . The decoder, the input weights and all RLS hyper-parameters never change.

Connectivity. Structural masks are nested: within a seed, edge (i, j) exists at density p iff $U_{ij} < p$, for one shared uniform matrix U (no self-connections). Non-zero weights are $W_{ij} = g G_{ij} / \sqrt{pN}$, with $g = 1.5$ and one shared Gaussian matrix G . All conditions of a seed share U , G , W_{in} , D , the training target order and initial states, and the held-out set. The densities are $p \in \{0.05, 0.10, 0.20, 0.40\}$.

Outcome. Held-out velocity NMSE under the fixed decoder, with learning switched off, on 30 trials (5 initial states $\times$ 6 targets). NMSE = 1 means “no better than predicting the mean velocity”. Lower is better.

Training length. Every network is trained for 200 trials and evaluated every 10 trials. The first 60 training trials and the held-out set are drawn exactly as in the original 60-trial runs; trials 61–200 come from a separate seeded stream. As a result, the trial-60 checkpoint of seeds 0–7 reproduces the original results to within 1.2×10^{-12} (all 224 overlapping checkpoints). The original snapshot is therefore a checkpoint of these runs, not a separate experiment.

2.2 Arms

Because the masks are nested, the control’s trainable *set* is identical across densities, not just its size. The random-subset arm keeps only the size: its trainable edges are spread over the dense mask, are not nested across densities, and start from the smaller dense-network weights $gG/\sqrt{pN}$. At $p = 0.05$ both equal-budget arms are the primary network (verified identical).

Table 1: Experimental arms. M_p is the structural mask at density p .

Arm	Trainable recurrent edges	Initial weights
All edges trainable (primary)	every existing edge ($\approx pN^2$)	$gG/\sqrt{pN}$ on M_p
Equal trainable budget (control)	the $p = 0.05$ edges $M_{0.05}$ in every density	as primary; edges in $M_p \setminus M_{0.05}$ are frozen
R2 gain-matched (both of the above)	as the parent arm	parent $W \times \text{scalar}$, so that $\rho(W_p) = \rho(W_{0.05})$ within the seed
Equal budget, random subset (B3)	a uniform random subset of M_p of the same size as $M_{0.05}$ (separate random stream)	as primary; the other edges of M_p are frozen
R3 frozen-drive factorial	$M_{0.05}$	see Section 2.3
R3 with unmatched total gain (B4)	$M_{0.05}$	see Section 2.3

2.3 R3: separating frozen drive from density

In the control, raising p changes three things at once: (1) the number of frozen edges, which is density itself; (2) the share of recurrent input variance carried by frozen edges, $f_p = (p-0.05)/p = 0, 0.5, 0.75$ or 0.875 ; and (3) correspondingly, the size of the trainable weights. To separate the first from the other two we build, for structural density p and frozen share f ,

$$W = \frac{g\sqrt{1-f}}{\sqrt{0.05N}} G \odot M_{0.05} + \frac{g\sqrt{f}}{\sqrt{(p-0.05)N}} G \odot (M_p \setminus M_{0.05}), \quad (1)$$

with only $M_{0.05}$ trainable. The expected total recurrent input variance per unit is g^2 in every cell.

Grid. We run the full factorial $p \in \{0.10, 0.20, 0.40\} \times f \in \{0.5, 0.75, 0.875\}$, plus $f = 0$ (identical for every p). Along the *density* axis at fixed f , the trainable sub-network is bit-for-bit identical across p and the frozen drive has the same variance; the only difference is how many frozen edges carry that variance. Along the *frozen-share* axis at fixed p , the edge set is identical; the only difference is how much drive is frozen.

Anchors. The diagonal cells $(0.10, 0.5)$, $(0.20, 0.75)$ and $(0.40, 0.875)$ are exactly the control networks, and $f = 0$ is exactly the $p = 0.05$ network. All four reproduced the control’s held-out NMSE exactly in every seed, so the factorial is anchored to the control rather than being a new model.

Avoiding a degenerate design. The obvious alternative, deleting the frozen edges, makes every network identical to $p = 0.05$ and cannot measure anything. Holding g^2 fixed also keeps “more frozen input” from being confounded with “more total gain”.

Unmatched-gain variant (B4). With g^2 fixed, more frozen drive also means smaller trainable weights (factor $\sqrt{1-f}$). To separate the two we also run the same 3×3 grid with the trainable weights left at the $p = 0.05$ scale, $g/\sqrt{0.05N}$, and the frozen weights as above. The total input variance is then $g^2(1+f)$; on the diagonal both edge sets keep their original scales, and $f = 0$ is again the $p = 0.05$ network. This variant fixes the trainable weights but trades the confound for another one: total gain (initial spectral radius 1.9–2.2 against 1.56) grows with f .

2.4 Contrasts and statistics

Contrasts. Per seed, $H1 = \text{NMSE}(0.05) - \frac{1}{3} \sum_{p>0.05} \text{NMSE}(p)$ (positive: denser is better) and $H2 = [\text{NMSE}(0.05) - \text{NMSE}(0.20)] - [\text{NMSE}(0.20) - \text{NMSE}(0.40)]$ (positive: diminishing returns). The primary-minus-control change is the per-seed difference of the two arms’ H1; it is possible because both arms share all seed-level randomness. For R3 we compute density $_f = \text{NMSE}(p=0.10, f) - \text{mean NMSE}(p=0.20, 0.40; f)$, frozen $_p = \text{NMSE}(f=0) - \text{mean NMSE}(p; f \geq 0.5)$, and per-seed least-squares slopes of NMSE on $\log_2 p$ and on f over the 3×3 factorial.

Inference. The network seed is the replicate. We report the mean, a seed-bootstrap 95% percentile interval of the mean (100 000 resamples, fixed generator seed), the number of positive seeds, and an exact two-sided sign test. The smallest attainable sign-test p is 0.0078 with 8 seeds and 3.1×10^{-5} with 16. No correction for multiple comparisons is applied; the intervals are descriptive.

Provenance. H1 and H2 were first committed together with the primary results (commit 3021081, 14 Jul 2026), so they are *not* preregistered, although early project documents called them “confirmatory”. They were fixed before the equal-budget control was run (920ff92, 29 Aug 2026). The robustness checks R1–R5 were specified in writing before they were run (9 Oct 2026), in response to an independent audit. We treat every contrast as exploratory.

Robustness checks. R1 (convergence): 200 trials; checkpoints 60, 100, 150 and 200. R2 (gain matching): initial spectral radius matched to the $p = 0.05$ network. R3 (frozen drive vs density): Section 2.3. R4 (network size): $N \in \{100, 200, 400\}$. R5 (more seeds): 16 seeds at $N = 200$. A second session (10 Oct 2026) extended R2 and R3 to 16 seeds and added B3, the random-subset equal budget (8 seeds), and B4, the unmatched-gain R3 variant (8 seeds); these were listed as pending checks in the first session’s record before they were run. Unless stated otherwise, everything uses $N = 200$; R2, R3 and R5 use 16 seeds and the other checks seeds 0–7.

3 Results

3.1 With all edges trainable, denser is better, robustly

Table 2: Mean held-out NMSE, $N = 200$, 16 seeds.

p	All trainable		Equal budget	
	trial 60	trial 200	trial 60	trial 200
0.05	0.447	0.320	0.447	0.320
0.10	0.359	0.207	0.497	0.359
0.20	0.271	0.155	0.475	0.379
0.40	0.141	0.077	0.492	0.386

With all edges trainable, H1 is positive at every checkpoint and in every variant at $N = 200$, and in every seed except one gain-matched seed (Table 2): +0.190 [0.154, 0.229] with 16 seeds at trial 60; +0.173 [0.135, 0.210] at trial 200; and +0.175 [0.128, 0.218], positive in 15/16, gain-matched at trial 200.

H2 (diminishing returns) is not supported at the original snapshot: +0.046 [−0.018, 0.111], 9/16. At trial 200 it is weakly positive (+0.087 [0.004, 0.170], 9/16), so we do not treat it as established.

3.2 Under an equal budget the advantage disappears

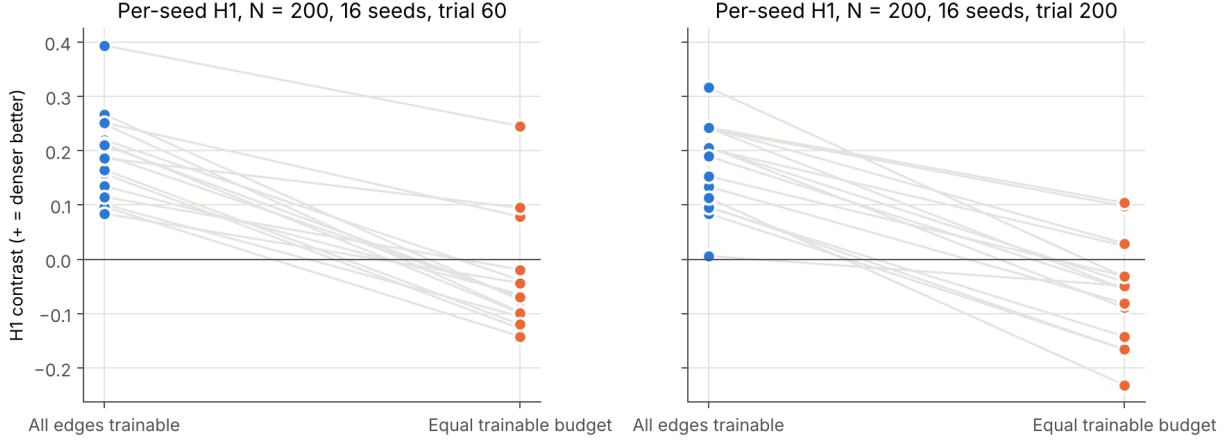


Figure 1: Per-seed H1 ($N = 200$, 16 seeds) with all edges trainable and under an equal trainable budget, at trial 60 (left) and trial 200 (right). Lines join the two arms of the same seed. Positive H1 means denser is better.

The paired primary-minus-control change in H1 is the most robust number in this study. It is positive in every seed of every check that ran (Table 3, Figure 1).

Table 3: Paired change in H1 (primary – control): mean, 95% seed-bootstrap CI, positive seeds.

Check	Paired change	Positive
Original snapshot, 8 seeds, trial 60	+0.267 [0.232, 0.304]	8/8
R1, trial 200	+0.261 [0.228, 0.296]	8/8
R2 gain-matched, 16 seeds, trial 60	+0.205 [0.169, 0.242]	16/16
R2 gain-matched, 16 seeds, trial 200	+0.214 [0.184, 0.249]	16/16
B3 random-subset budget, trial 60	+0.246 [0.170, 0.317]	8/8
B3 random-subset budget, trial 200	+0.248 [0.207, 0.285]	8/8
R5, 16 seeds, trial 60	+0.231 [0.193, 0.267]	16/16 ($p = 3 \times 10^{-5}$)
R5, 16 seeds, trial 200	+0.229 [0.191, 0.265]	16/16
R4, $N = 100$, trial 200	+0.179 [0.096, 0.273]	8/8
R4, $N = 400$, trial 200	+0.069 [0.060, 0.079]	8/8

The denser networks’ advantage therefore depends on their extra trainable edges. Holding the number of trainable edges fixed removes all of it, whether the trainable edges are the sparse network’s edges or a random subset of the dense mask (Figure 2).

3.3 The sign reversal is small and fragile

The point estimate is negative in every check (Table 4), but the effect is small and fragile. Its magnitude roughly halves when the seed count doubles: seeds 8–15 alone average +0.008 at trial 60 (−0.049 at trial 200), so the original 8-seed estimate was optimistic. Under gain matching with 16 seeds the interval includes zero at both checkpoints, and only 9–10 of 16 seeds are negative. The random-subset budget (B3) gives a negative H1 at 8 seeds, but so did the original control at

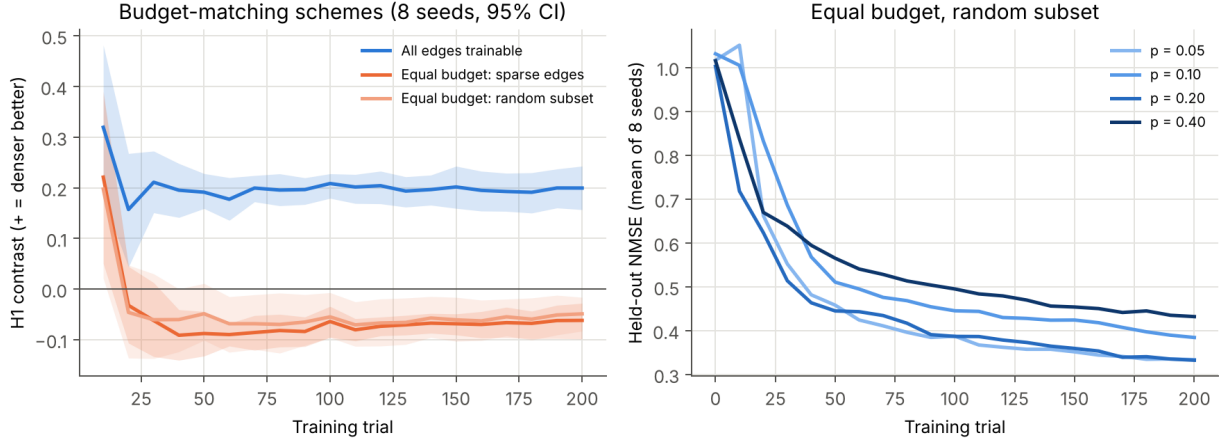


Figure 2: Budget-matching schemes (8 seeds). Left: H1 over training with all edges trainable, with the equal budget formed by the sparse network’s edges, and with the equal budget formed by a random subset of each dense mask (B3); shaded bands are 95% seed-bootstrap CIs. Right: mean held-out NMSE per density in the random-subset arm.

Table 4: Equal-budget H1 in each check (negative = sparse better).

Check	Equal-budget H1	Negative seeds
Original 8 seeds, trial 60	−0.090 [−0.113, −0.068]	8/8
R1, trial 100 / 150 / 200	−0.064 / −0.069 / −0.062, all CIs < 0	7/8, 8/8, 7/8
R2 gain-matched, 8 seeds, trial 60	−0.068 [−0.117, −0.024]	8/8
R2 gain-matched, 8 seeds, trial 200	−0.045 [−0.108, +0.009]	5/8
R2 gain-matched, 16 seeds, trial 60	−0.027 [−0.073, +0.019]	10/16
R2 gain-matched, 16 seeds, trial 200	−0.039 [−0.086, +0.004]	9/16
B3 random-subset budget, trial 60	−0.069 [−0.125, −0.015]	6/8
B3 random-subset budget, trial 200	−0.049 [−0.083, −0.017]	6/8
R5, 16 seeds, trial 60	−0.041 [−0.083, +0.011]	13/16
R5, 16 seeds, trial 200	−0.055 [−0.100, −0.011]	12/16
R4, $N = 100$, trial 200	−0.084 [−0.142, −0.023]	6/8
R4, $N = 400$, trial 200	−0.014 [−0.027, −0.002]	6/8

8 seeds before it halved. “Denser is worse under an equal budget” is therefore at most a modest tendency, not a clean reversal. We do not claim it.

Spectral radius. The initial spectral radius was not controlled in the original runs (1.48–1.65). In seeds 0–7 the sparse-minus-denser radius gap correlated with primary H1 at $r = -0.90$. Over 16 seeds this falls to $r = -0.26$ (primary) and $r = +0.20$ (control). Matching the radii (R2) leaves both the primary advantage and the paired change intact (Figure 3). The strong correlation was a small-sample coincidence, not a mechanism.

3.4 Convergence

The original trial-60 snapshot was taken mid-learning: 23/32 primary networks were still improving between trials 50 and 60. Training to 200 trials lowers every NMSE substantially, but every contrast

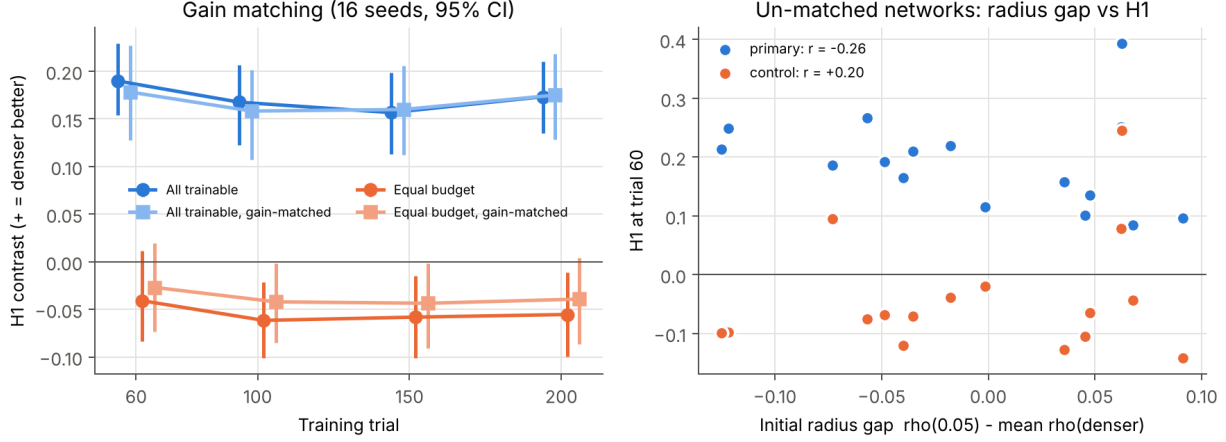


Figure 3: Gain matching (16 seeds). Left: H1 at each checkpoint with and without initial spectral-radius matching (95% CIs). Right: in the unmatched networks, the initial radius gap $\rho(0.05) - \text{mean } \rho(\text{denser})$ against H1 at trial 60.

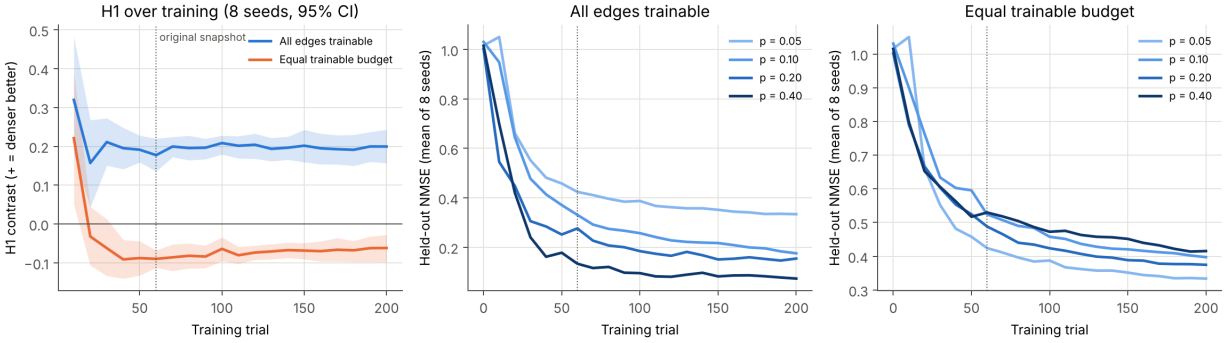


Figure 4: Convergence (8 seeds). Left: H1 over 200 training trials for both arms (95% CIs); the dotted line marks the original trial-60 snapshot. Middle and right: mean held-out NMSE per density in each arm.

stays within its trial-60 interval (Figure 4): primary H1 goes from +0.177 to +0.199 [0.157, 0.243]; control H1 from -0.090 to -0.062 [$-0.099, -0.028$]; and the paired change from +0.267 to +0.261.

The networks are still not fully converged at 200 trials. Between trials 150 and 200, 23/32 primary and 26/32 control networks improved (median relative NMSE change -11% and -7%). Between trials 190 and 200 only about half improved (17/32 and 15/32), which is consistent with a plateau plus noise. H1 was flat from trial 40 onward in both arms.

3.5 No density component is detected in the control's residual tendency

Density at a fixed frozen share has no detectable effect (R3, 16 seeds; Table 5, Figure 5).

All intervals include zero and are centred near it; the 16-seed slope intervals are about half as wide as the 8-seed ones. This is the most solid part of the R3 result, but it is an absence of evidence, not evidence of absence. The trial-200 density contrasts reach -0.08 to -0.09 , and the slope's upper bound ($+0.047$ per doubling, two doublings from $p = 0.10$ to 0.40) allows up to ≈ 0.09 NMSE. Both exceed the control's reversal (-0.055), so a density component of that size cannot be

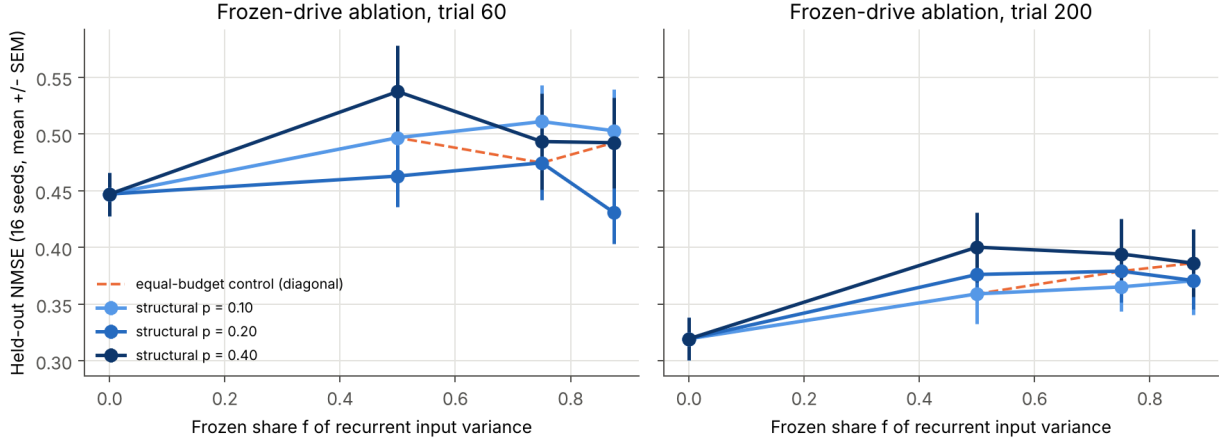


Figure 5: Frozen-drive ablation R3 (16 seeds, mean $\pm$ SEM), at trial 60 (left) and trial 200 (right). Each line is one structural density; the dashed line joins the diagonal cells, which are exactly the equal-budget control networks.

Table 5: R3 density contrasts (16 seeds).

R3 contrast	Trial 60	Trial 200
density, $f = 0.5$	-0.003 $[-0.069, 0.062]$	-0.029 $[-0.093, 0.042]$
density, $f = 0.75$	$+0.027$ $[-0.049, 0.104]$	-0.021 $[-0.077, 0.034]$
density, $f = 0.875$	$+0.041$ $[-0.038, 0.121]$	-0.008 $[-0.078, 0.065]$
slope per doubling of p	$+0.002$ $[-0.037, 0.043]$	$+0.014$ $[-0.018, 0.047]$

excluded. Excluding it would need intervals of about ± 0.03 , i.e. roughly four times as many seeds.

Adding frozen drive at a fixed density: a small cost, clear only at trial 200 (Table 6).

Table 6: R3 frozen-drive contrast: $\text{NMSE}(f = 0)$ minus mean NMSE over $f \geq 0.5$ (16 seeds); positive seeds in parentheses.

	Trial 60	Trial 200
$p = 0.10$	-0.056 $[-0.116, 0.003]$ (6/16)	-0.045 $[-0.092, 0.001]$ (6/16)
$p = 0.20$	-0.009 $[-0.067, 0.052]$ (7/16)	-0.056 $[-0.110, -0.001]$ (5/16)
$p = 0.40$	-0.061 $[-0.124, 0.007]$ (4/16)	-0.074 $[-0.129, -0.022]$ (4/16)

With 8 seeds two of these three intervals excluded zero at trial 60; with 16 seeds none does, and the per-seed sign tests are not significant ($p \geq 0.077$). At trial 200 the mean NMSE rises from 0.320 ($f = 0$) to 0.36–0.40 for every cell with $f \geq 0.5$, and the slope over $f \in [0.5, 0.875]$ is flat (-0.005 $[-0.058, 0.043]$).

What the unmatched-gain variant adds (B4, 8 seeds). Keeping the trainable weights at full size and adding frozen drive on top is far worse: NMSE 0.53–0.72 at trial 200 against 0.334 at $f = 0$ (frozen-drive cost -0.30 to -0.34 , 8/8 seeds in every column), and the cost now grows with f (slope $+0.350$ $[0.273, 0.426]$ per unit f , 8/8; Figure 6). Every unmatched cell is worse than

the matching gain-matched R3 cell (mean difference -0.260 $[-0.300, -0.222]$, 8/8). So shrinking the trainable weights is not what makes R3’s frozen-drive cells worse: at the same frozen drive, full-size trainable weights do worse, not better. Because the unmatched networks also have larger total gain, B4 cannot say whether frozen drive as such or the extra gain does the damage; density again has no effect at fixed f (slope $+0.007$ $[-0.024, 0.037]$).

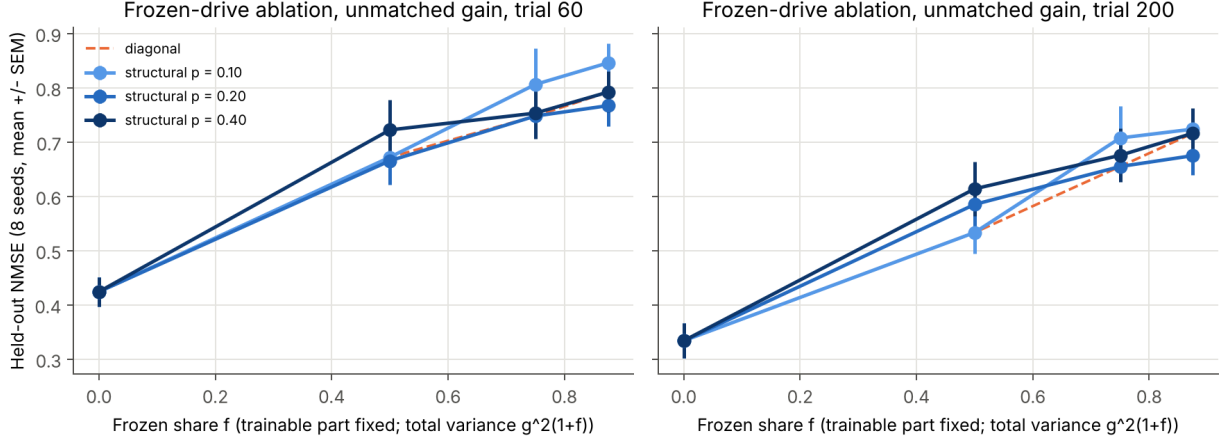


Figure 6: Frozen-drive ablation with unmatched gain (B4; 8 seeds, mean $\pm$ SEM). The trainable weights keep the $p = 0.05$ scale, so total input variance is $g^2(1 + f)$.

Interpretation. The equal-budget control, rebuilt from the factorial’s diagonal, compares a network with no frozen drive ($p = 0.05$) against three networks that have some. Density at fixed frozen share contributes nothing detectable, although the ablation is not precise enough to exclude a density component as large as the control’s residual negative H1. Attributing it to frozen recurrent drive is consistent with the R3 step at trial 200 and with B4, but the 16-seed evidence for that step is weak (trial 200 only, sign tests not significant), so we state it as the most likely explanation rather than as an established result.

3.6 Network size

$N = 100$. Learning is poor: NMSE 0.60–0.79 at trial 200 and 0.75–0.85 at trial 60, against a chance level of 1. The primary H1 does not differ from zero (trial 60: $+0.007$ $[-0.104, 0.118]$; trial 200: $+0.095$ $[-0.029, 0.216]$). This matches the earlier 3-seed $N = 100$ pilot, which had the opposite sign and which we now read as a regime in which too little is learned for density to matter. The control H1 is negative at trial 200 (-0.084 $[-0.142, -0.023]$). The paired change is $+0.179$ $[0.096, 0.273]$, 8/8.

$N = 400$. Learning is much better (trial-200 NMSE 0.012–0.080 with all edges trainable), so every contrast is smaller in absolute NMSE units. The pattern of $N = 200$ is reproduced at both checkpoints (Figure 7): primary H1 is $+0.062$ $[0.043, 0.077]$ at trial 60 and $+0.055$ $[0.040, 0.068]$ at trial 200, 8/8 positive; equal-budget H1 is -0.025 $[-0.042, -0.008]$ and -0.014 $[-0.027, -0.002]$, 6/8 negative; the paired change is $+0.086$ $[0.077, 0.097]$ and $+0.069$ $[0.060, 0.079]$, 8/8. In relative terms the effect is large. With all edges trainable, $p = 0.40$ reaches 15% of the $p = 0.05$ NMSE (0.012 vs 0.080). Under the equal budget, $p = 0.05$ is still the best condition (0.080 vs 0.084–0.103).

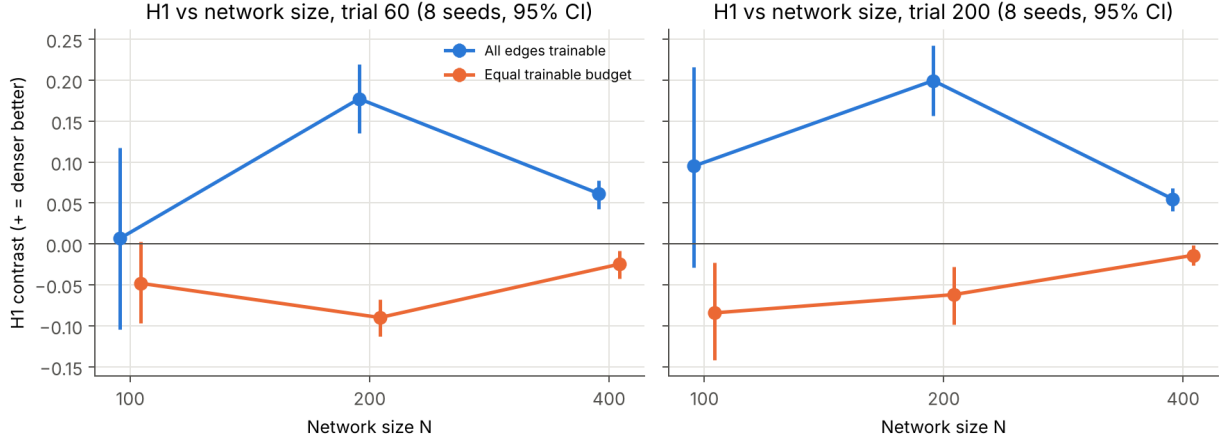


Figure 7: H1 against network size N at trial 60 (left) and trial 200 (right), 8 seeds, 95% CIs.

Across sizes. The sign pattern holds wherever the networks learn. Absolute H1 is not comparable across N , because the NMSE floor differs.

4 Limitations

- **One model family and one task.** Rate units with tanh, FORCE/RLS on recurrent weights, a fixed random decoder, and six-target reaching with constant velocities. We do not test other learning rules (BPTT, node perturbation), output training, or structured connectivity, so our results do not speak directly to the sparse-advantage findings obtained with other architectures and training procedures [4, 5].
- **Small samples.** 16 seeds for R2, R3 and R5; 8 seeds for R1, R4, B3 and B4. R5 and R2 show that 8-seed effect sizes in this system can shrink by half; the $N = 400$, random-subset and unmatched-gain results rest on 8 seeds.
- **Not fully converged.** At 200 trials, roughly half of the networks still improve over the last 10 trials. Contrasts were stable from trial ~ 40 onward, but asymptotic behaviour is not established.
- **Budget-matching schemes.** We tested two: the sparse network’s edges and a random subset of each dense mask (8 seeds). Both remove the advantage. Other schemes, such as an equal number of trainable weights per neuron, were not tested.
- **Frozen drive vs gain.** R3 holds total variance at g^2 , so more frozen drive also means smaller trainable weights; B4 keeps the trainable weights fixed but lets total gain grow. Together they rule out “smaller trainable weights” as the source of R3’s cost, but they do not separate frozen drive from total gain. The missing cell is shrunken trainable weights with the frozen edges deleted.
- **Gain matching rescales all weights.** It matches the spectral radius but changes the trainable weights’ scale too.

- **No population-geometry analysis.** We report task error only. In a six-target task the dimensionality of population activity is likely to be bounded by task complexity rather than by the circuit [7], so we do not interpret manifold dimension here.
- **Analysis choices.** H1/H2 are exploratory, not preregistered (Section 2.4). No multiplicity correction is applied across the ~ 60 reported intervals.
- **Biology.** No claim about cortical connectivity follows from this model.

5 Conclusion

In a FORCE-trained motor RNN, the performance advantage of denser recurrent connectivity is explained by the number of trainable recurrent edges. Holding the trainable set fixed removes it in 16/16 seeds. This holds across training length, spectral-radius matching (16/16) and a second budget-matching scheme (random subset, 8/8). It holds at $N = 400$, and at $N = 100$ after 200 trials; at $N = 100$ the networks barely learn, and the all-trainable advantage itself is not significant there.

Under a matched budget, denser networks are no better and possibly slightly worse. That residual handicap is small and does not survive gain matching with 16 seeds. A factorial ablation detects no density component in it, but cannot exclude one of the same size. It is most plausibly a cost of the frozen recurrent drive the control introduces, but that attribution also rests on weak evidence.

Studies that vary connectivity under recurrent learning should report the trainable-parameter count as a separate factor. They should also note that “freezing the extra edges” is not a neutral control.

Code and data availability

All code, per-network results and analysis outputs are at <https://github.com/Thom-320/nma-motor-rnn-connectivity> (branch preprint-2026-10). Every number in this note is regenerated by `scripts/run_robustness_grid.py` and `scripts/analyze_robustness.py`; per-network CSVs are in `results/robustness/raw/` and contrasts in `results/robustness/analysis/`. Code is released under the BSD-3-Clause licence.

Third-party material

The network model, task and learning rule follow Feulner & Clopath [1]; their code (<https://github.com/babaf/neural-manifold-and-plasticity>) is distributed under the MIT licence. The project started from the Neuromatch Academy Computational Neuroscience motor-RNN project template and a teaching fork of the Feulner & Clopath code by Steeve Laquitaine; Neuromatch Academy course materials are licensed CC BY 4.0 (text) and BSD-3-Clause (code). Sources and licences are listed in `THIRD_PARTY.md` in the repository. This work is not a new learning algorithm.

Use of AI tools

Claude Code (Anthropic), an AI coding assistant, helped write and test the simulation and analysis code and helped draft and edit this text. Thomas Chisica checked the reported numbers against

the analysis outputs and takes responsibility for the content. The AI tool is not an author.

Author contributions

T.C. designed the equal-budget control and the robustness checks, ran the simulations and analyses, and wrote the manuscript. The primary all-edges-trainable experiment was developed within a Neuromatch Academy team project (see Acknowledgements).

Competing interests

The author declares no competing interests.

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References

- [1] Barbara Feulner and Claudia Clopath. Neural manifold under plasticity in a goal driven learning behaviour. *PLOS Computational Biology*, 17(2):e1008621, 2021. doi: 10.1371/journal.pcbi.1008621.
- [2] Emil Wörnberg and Arvind Kumar. Perturbing low dimensional activity manifolds in spiking neuronal networks. *PLOS Computational Biology*, 15(5):e1007074, 2019. doi: 10.1371/journal.pcbi.1007074.
- [3] Herbert Jaeger and Harald Haas. Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication. *Science*, 304(5667):78–80, 2004. doi: 10.1126/science.1091277.
- [4] Mikail Khona, Sarthak Chandra, Joy J. Ma, and Ila R. Fiete. Winning the lottery with neural connectivity constraints: Faster learning across cognitive tasks with spatially constrained sparse RNNs. *Neural Computation*, 35(11):1850–1869, 2023. doi: 10.1162/neco_a_01613.
- [5] Rieke Fruengel and Marcel Oberlaender. Sparse connectivity enables efficient information processing in cortex-like artificial neural networks. *Frontiers in Neural Circuits*, 19:1528309, 2025. doi: 10.3389/fncir.2025.1528309.
- [6] David Sussillo and L. F. Abbott. Generating coherent patterns of activity from chaotic neural networks. *Neuron*, 63(4):544–557, 2009. doi: 10.1016/j.neuron.2009.07.018.
- [7] Peiran Gao, Eric Trautmann, Byron M. Yu, Gopal Santhanam, Stephen Ryu, Krishna V. Shenoy, and Surya Ganguli. A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv*, 2017. doi: 10.1101/214262. Preprint 214262.